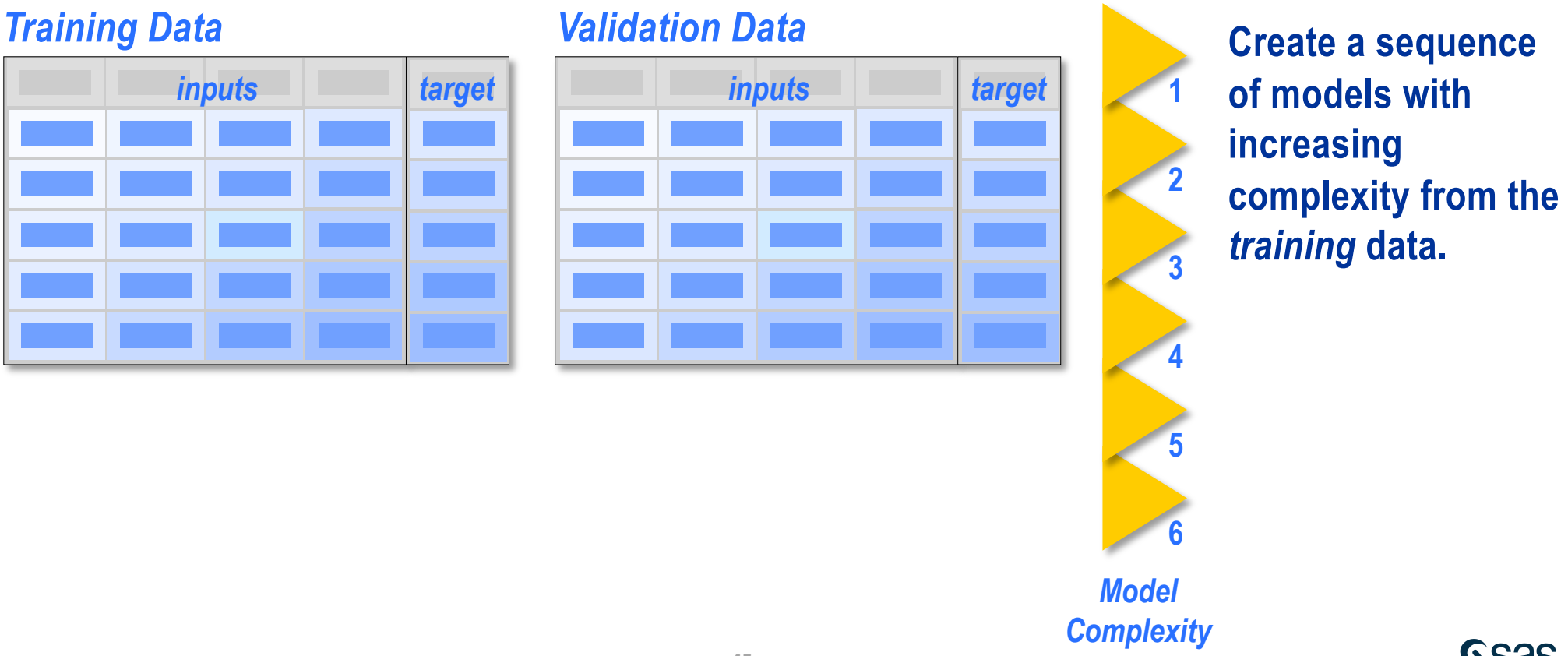
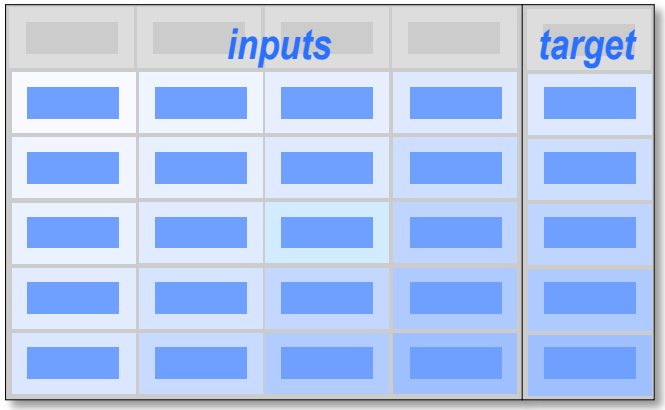
Predictive Model Sequence



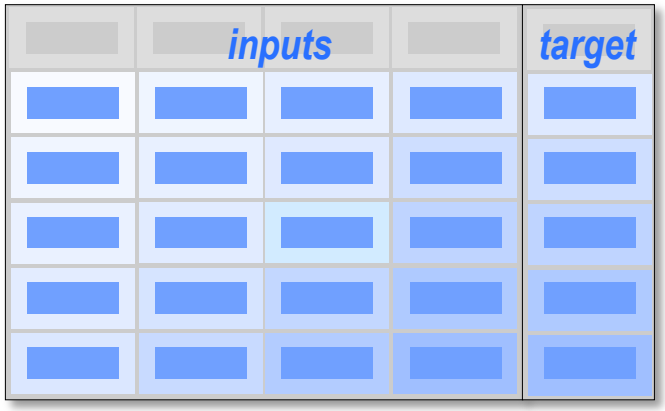
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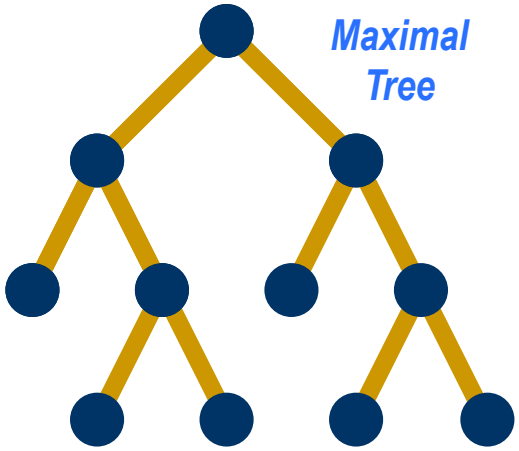
Maximal Tree

Training Data



Validation Data





Maximal Tree

A maximal tree is the most complex model in the sequence.

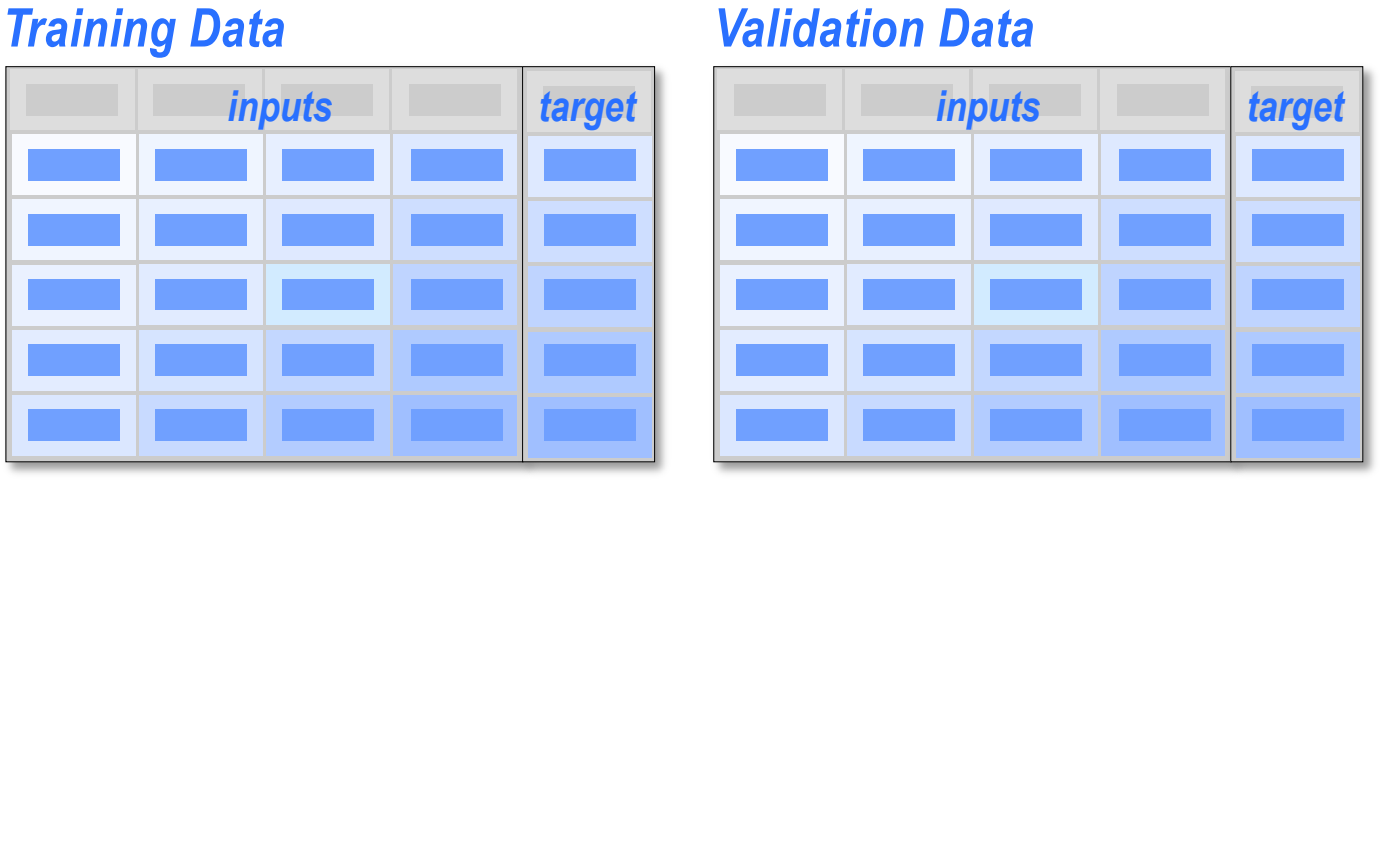


Model Complexity

Create a sequence of models with increasing complexity from the *training* data.

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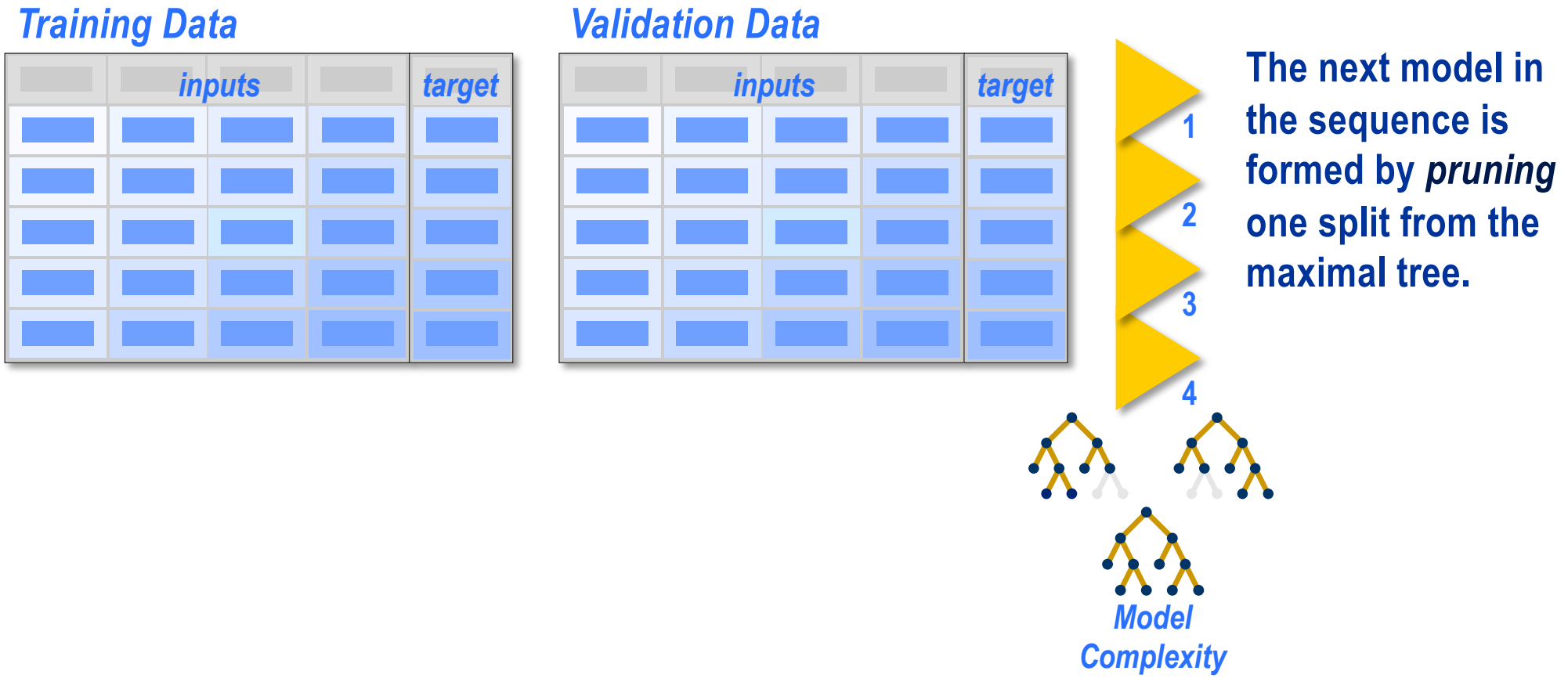
Maximal Tree



A maximal tree is the most complex model in the sequence.

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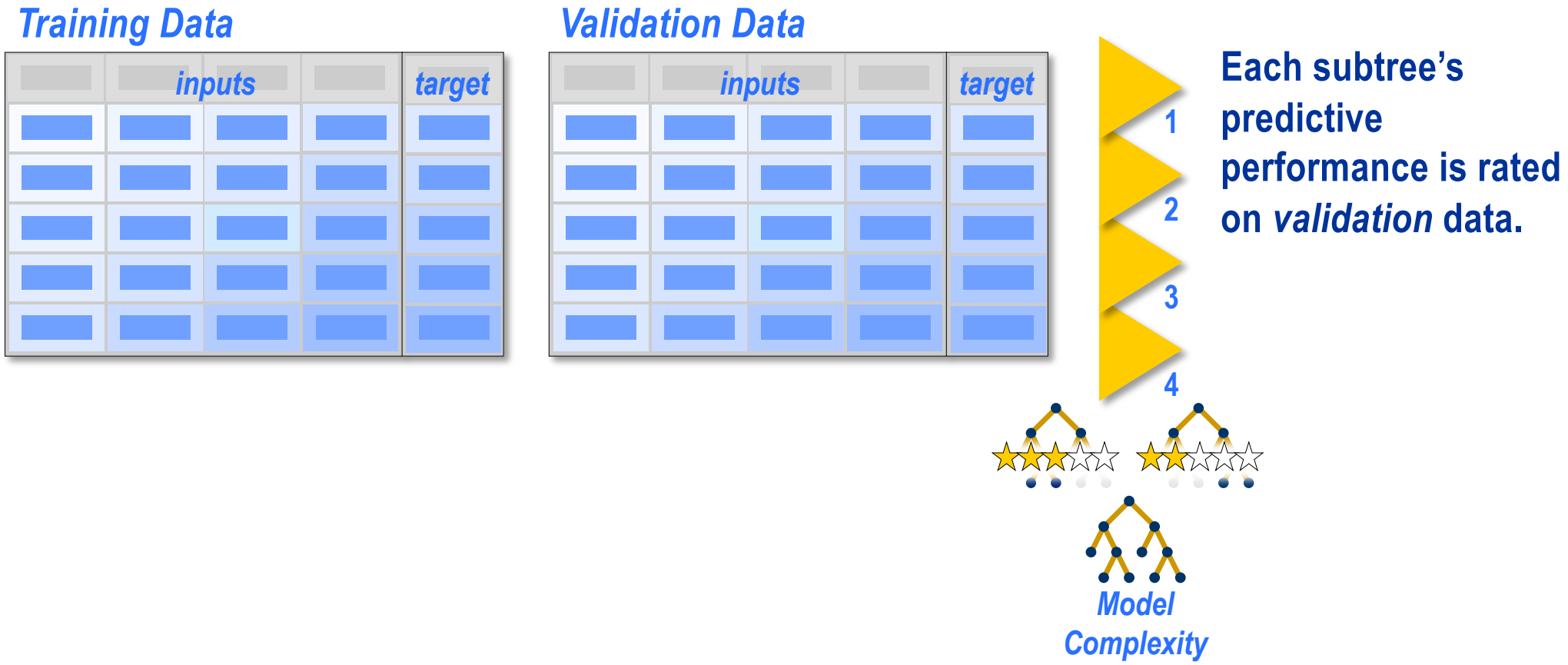
Pruning One Split



The next model in the sequence is formed by *pruning* one split from the maximal tree.

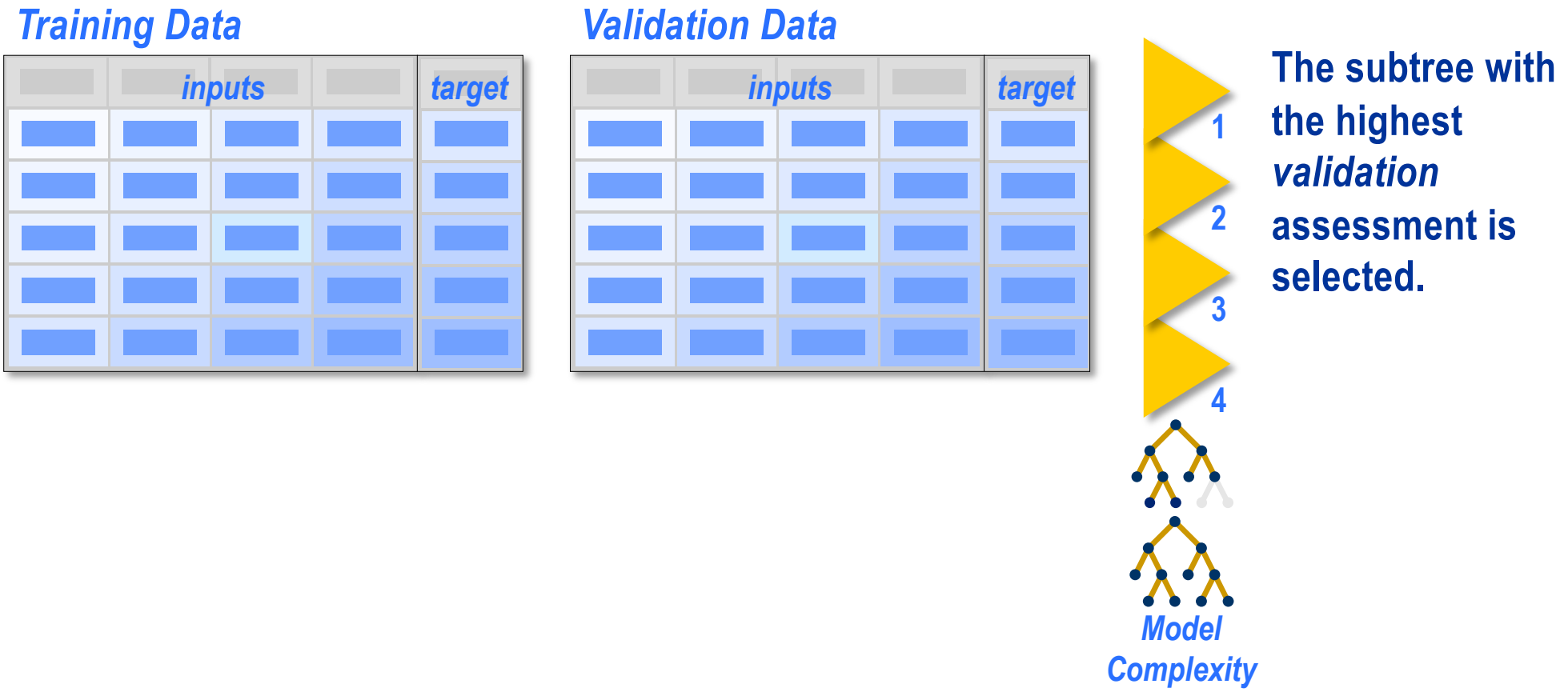
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Pruning One Split



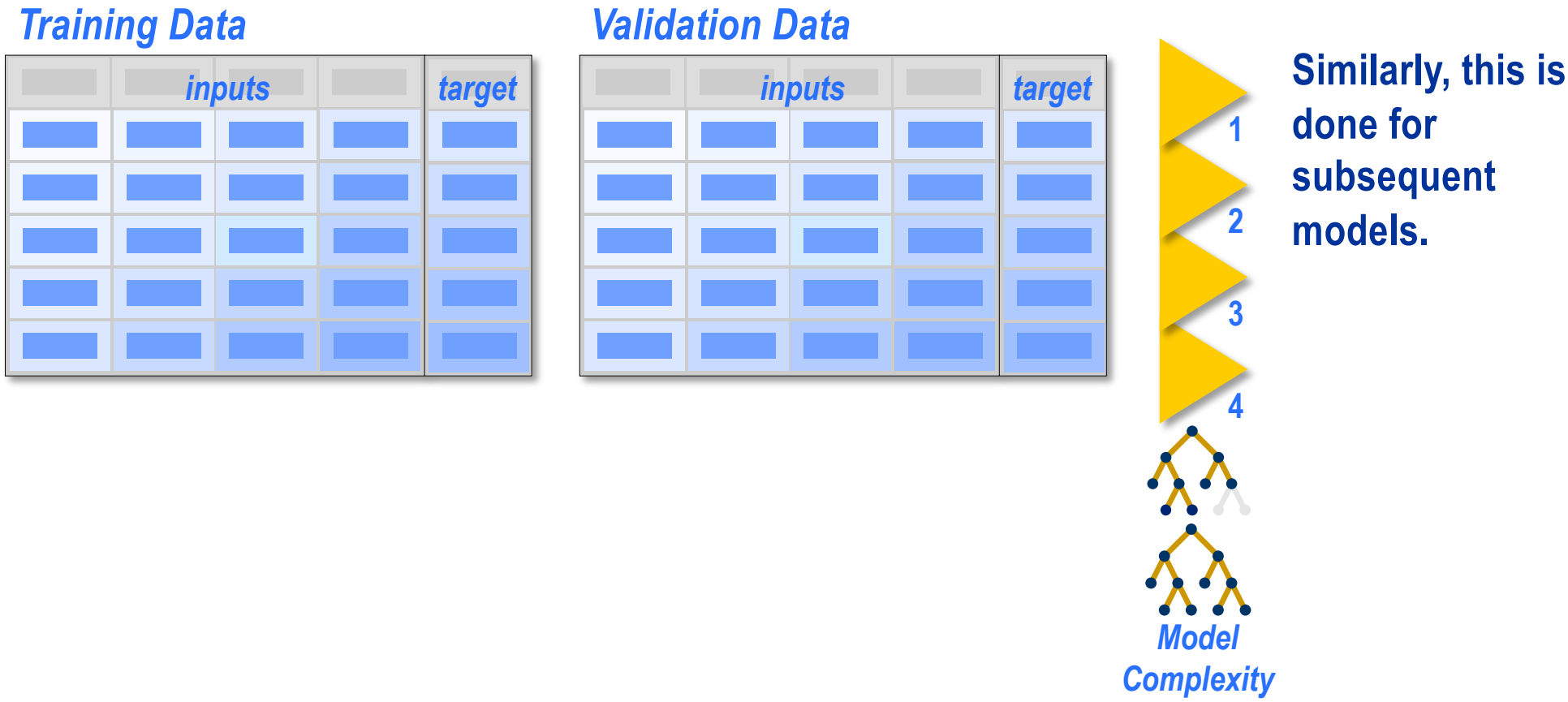
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Pruning One Split



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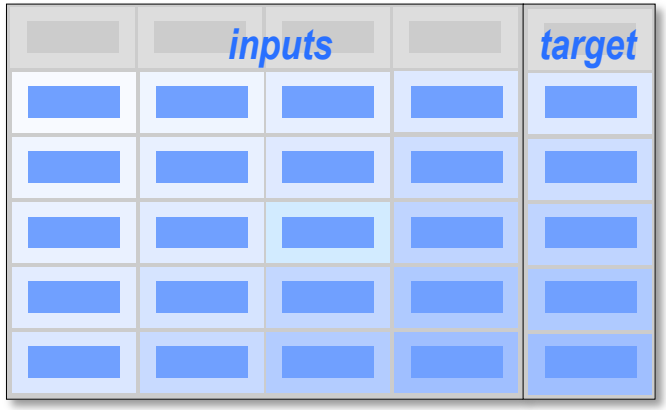
Pruning Two Splits



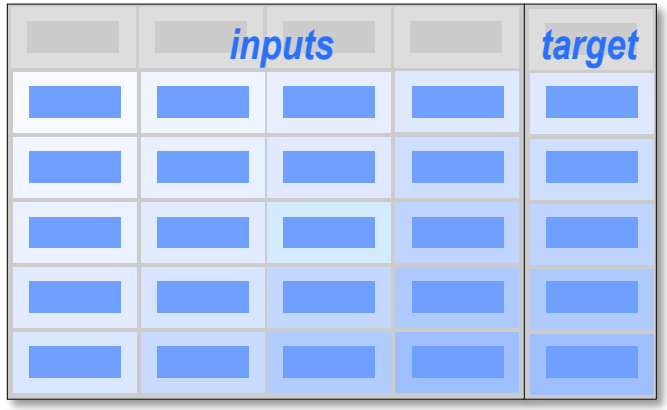
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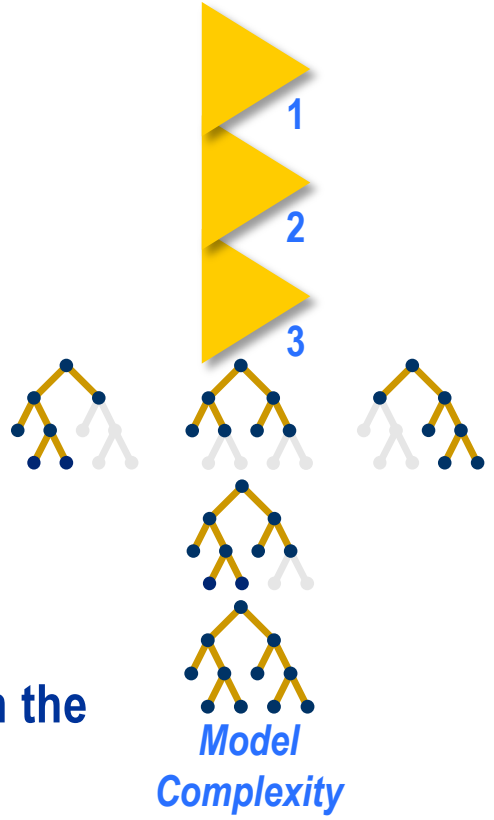
Pruning Two Splits

Training Data



Validation Data



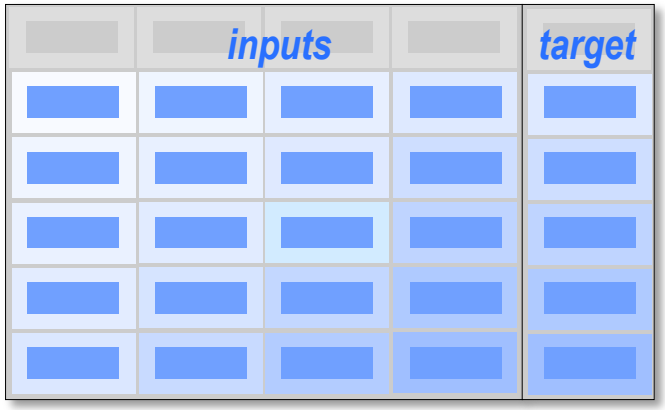


Prune two splits from the maximal tree,…

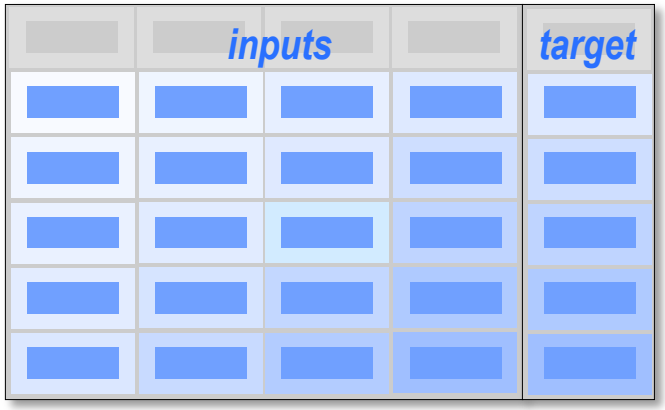
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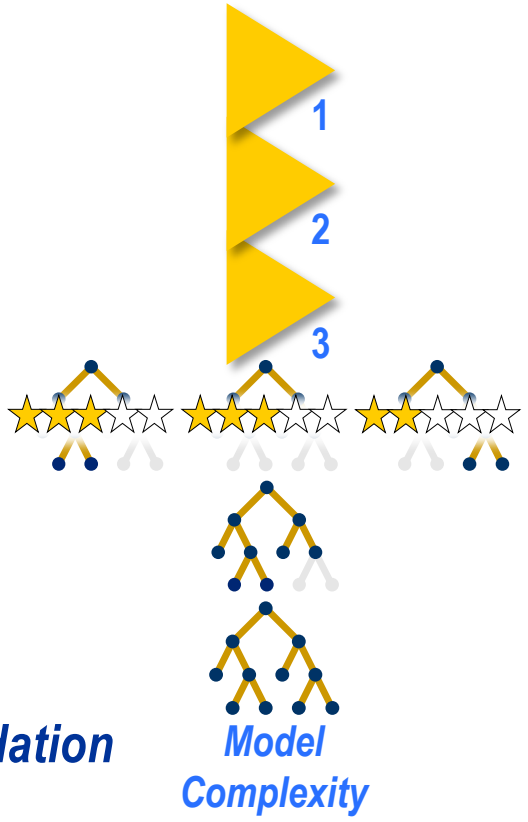
Pruning Two Splits

Training Data



Validation Data





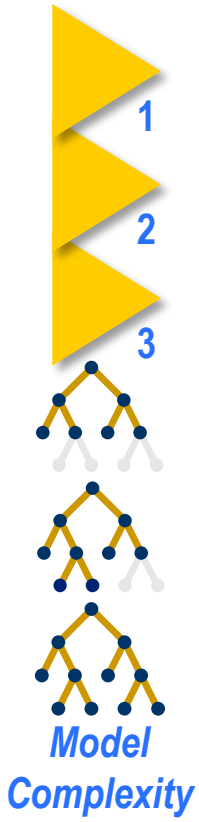
…rate each subtree using *validation* assessment, and…

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Pruning Two Splits



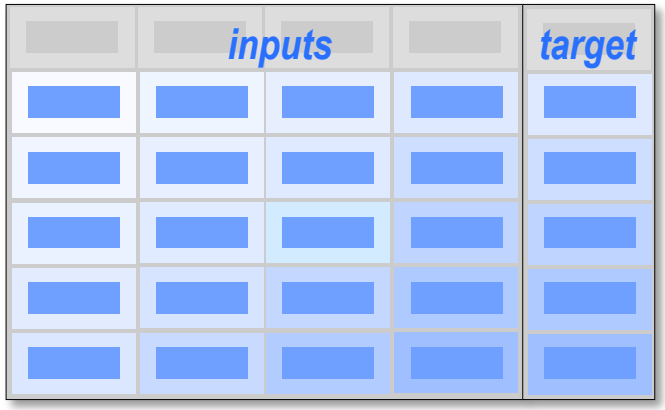
…select the subtree with the best assessment rating.



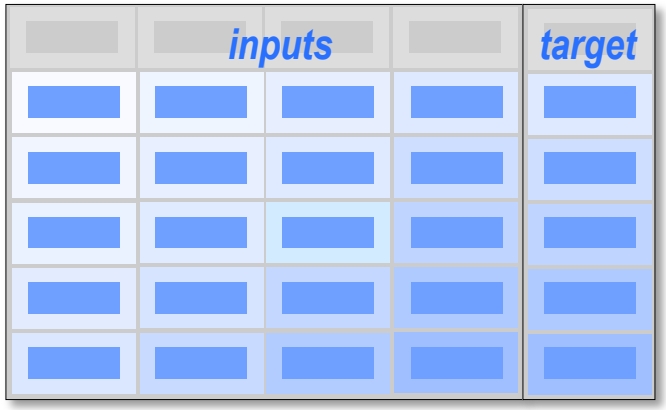
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Subsequent Pruning

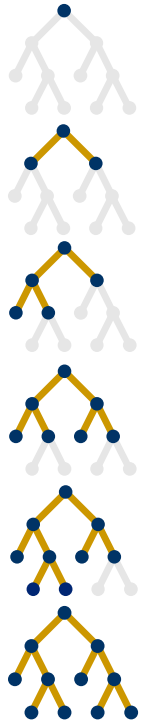
Training Data



Validation Data



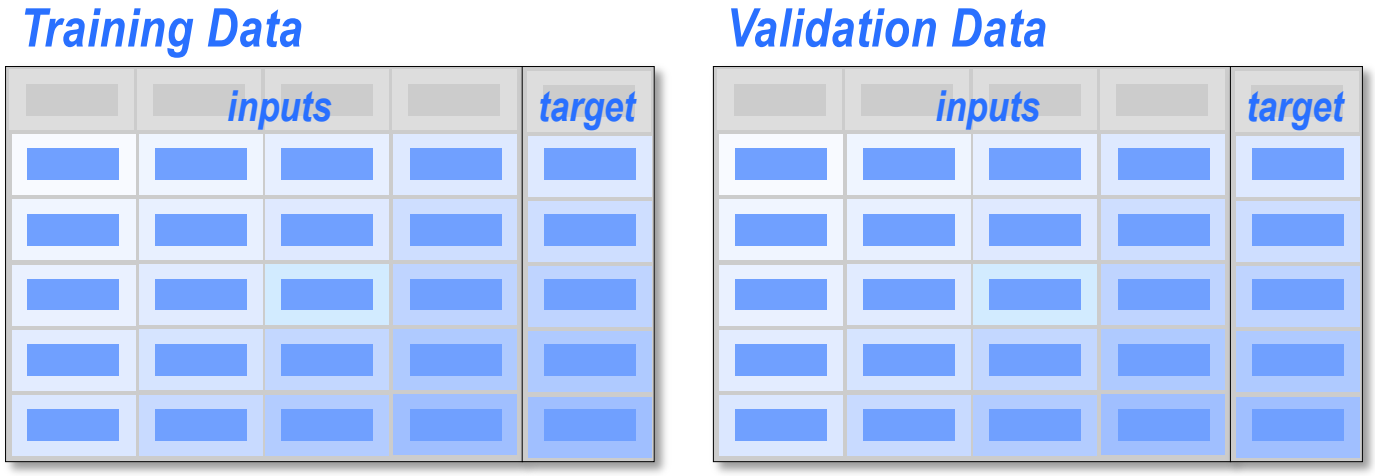
Continue pruning until all subtrees are considered.



Model Complexity

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Selecting the Best Tree

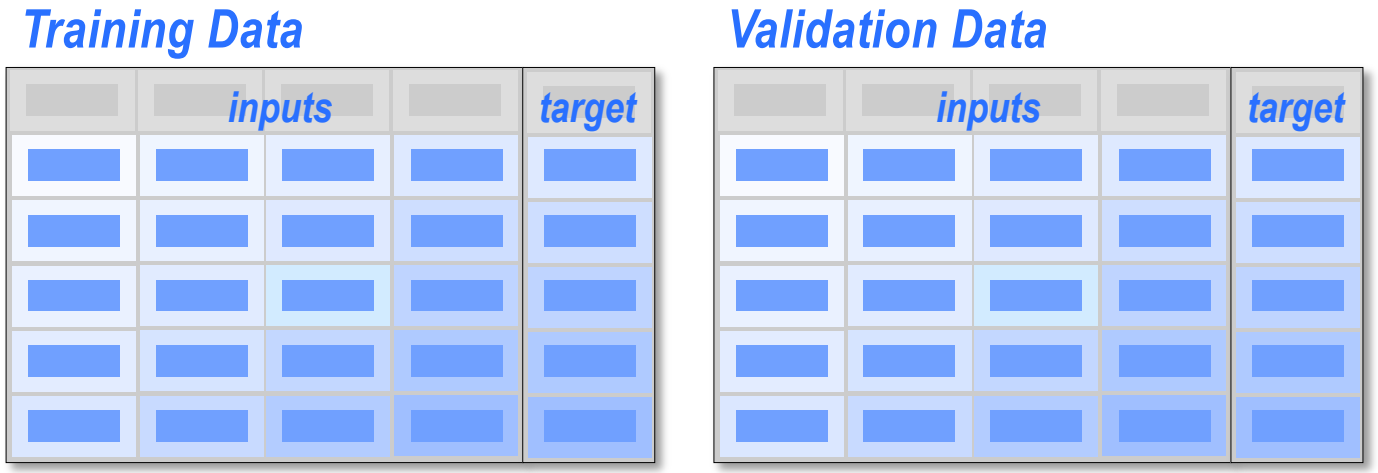


Compare *validation* assessment between tree complexities.



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Validation Assessment

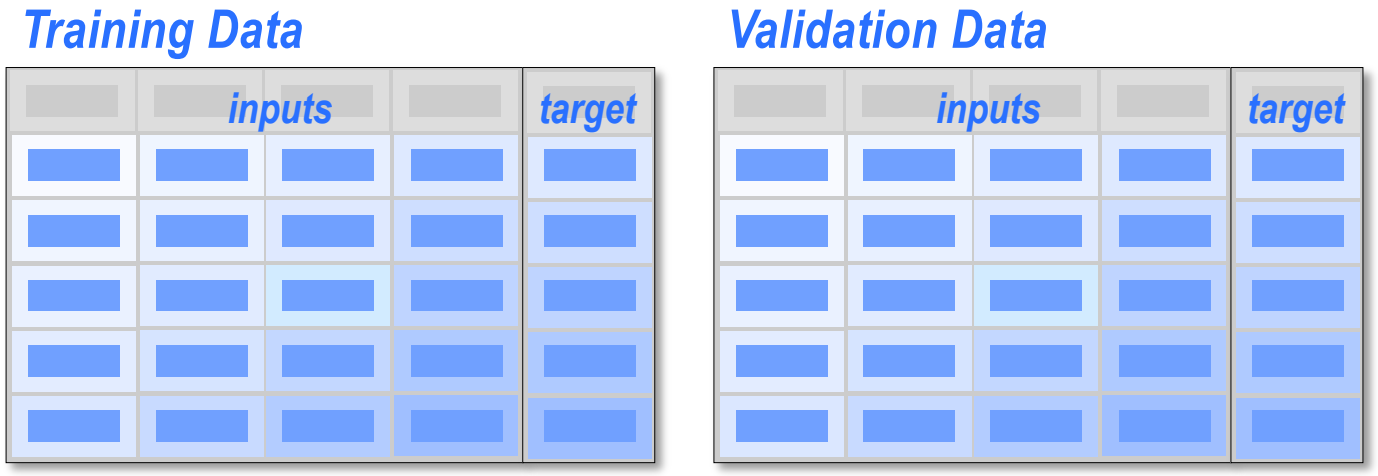




Choose the simplest model with the highest *validation* assessment.

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Validation Assessment



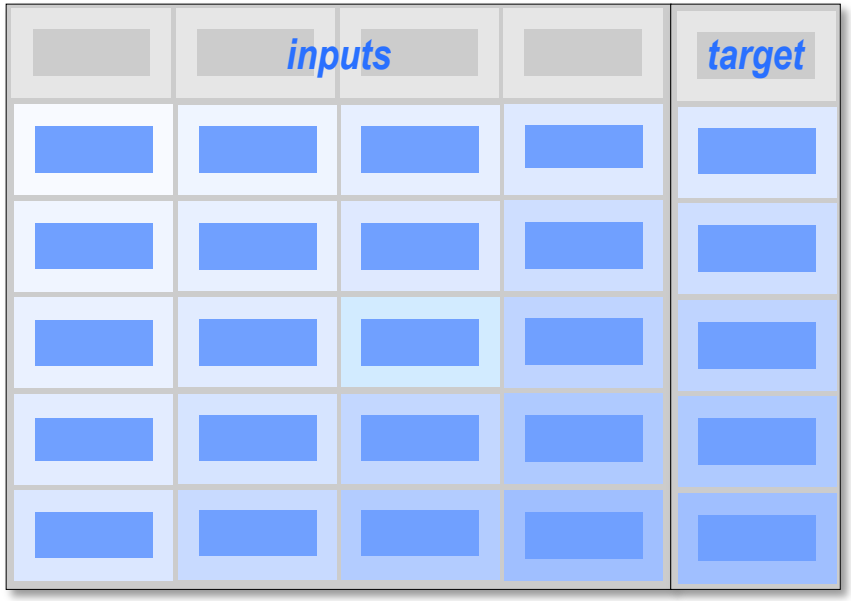


What are appropriate validation assessment ratings?

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Assessment Statistics

Validation Data



Ratings depend on…

- target measurement (binary, continuous, and so on)
- prediction type (decisions, rankings, estimates)

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